

MULTIVIEW: A Portable C99 System for Two-Camera Scene Analysis and 3D Model Extraction

A self-contained overview of the system and its mathematics

Project documentation — v0.2.0

August 2, 2026

Abstract

MULTIVIEW is a small, dependency-free C99 library for metric scene analysis and 3D model extraction from the synchronized video streams of *two stationary cameras* observing an overlapping volume, optionally augmented by a collection of still photographs and, in a later stage, range-finder measurements. Calibration is part of the system: the rig is calibrated from views of a printed letter-page checkerboard of known square size, or from a computer display under our control showing synchronized Gray-code patterns. This document doubles as a self-study overview of the underlying mathematics: each concept is developed from first principles, cross-linked to the modules that implement it, illustrated by a *hand-computable* worked example, and sourced in the bibliography. All quantitative claims ([section 13](#)) are reproducible from the deterministic test programs in the repository.

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1 Introduction and capabilities

The target deployment is deliberately narrow and fully under our control: two cameras are rigidly mounted and observe the same working volume. The system first *calibrates itself* from views of a known planar target ([section 5](#)); thereafter video is processed frame by frame. Because the cameras do not move, every geometric quantity that depends only on the rig — projection matrices, the fundamental matrix, the rectifying homographies — is computed *once* and reused for every frame, which is the central efficiency win of the stationary-rig assumption.

The implemented core (v0.2.0) provides:

1. plane-based calibration of intrinsics, distortion, and pose from a printed letter-page checker-board or from a controlled display showing Gray-code patterns ([section 5](#));
2. a calibrated pinhole camera model with Brown radial–tangential lens distortion, forward projection and ray back-projection ([section 4](#));
3. exact two-view epipolar geometry from the calibration, and the normalized 8-point algorithm to estimate or verify it from image correspondences ([section 6](#));
4. N -view linear triangulation of corresponding image points into metric 3D, with reprojection-error reporting ([section 7](#));
5. planar stereo rectification of the pair, and dense correspondence by block matching with sub-pixel refinement, giving per-pixel depth ([section 8](#));
6. point-cloud assembly and PLY export for inspection in standard mesh tools, plus minimal PGM image I/O ([section 9](#)).

Planned extensions — structure-from-motion over an auxiliary photo collection and fusion of range-finder data — are designed but not yet implemented; their theory and integration points are described in [section 11](#) so that the code can grow into them without rearchitecting.

How to read this document. [section 3](#) collects the small amount of linear algebra everything else stands on. [section 4–section 8](#) then follow the data through the system: camera model, calibration, two-view geometry, triangulation, dense stereo. Every section ends in a worked example with numbers chosen so the computation fits on paper; doing them by hand *is* the intended self-study exercise, and each can then be checked against the library.

Design constraints. The library is strict C99 with no dependencies beyond `libm`, row-major double arrays throughout, and fully deterministic (no clocks, no ambient randomness; the demos use fixed-seed generators). This keeps it portable from desktop to embedded targets, and keeps every experiment repeatable — a prerequisite for the controlled-environment, learning-oriented use the project is intended for.

2 System overview

Module	Header	Responsibility
<code>mat</code>	<code>mv/mat.h</code>	small dense linear algebra, Jacobi SVD, null vectors
<code>cam</code>	<code>mv/cam.h</code>	camera model, projection, rays, distortion
<code>target</code>	<code>mv/target.h</code>	printed checkerboard model + printable rendering
<code>graycode</code>	<code>mv/graycode.h</code>	display-based active target (Gray-code patterns)
<code>calib</code>	<code>mv/calib.h</code>	homographies; Zhang calibration; radial estimate
<code>epipolar</code>	<code>mv/epipolar.h</code>	F from calibration; normalized 8-point; E
<code>triangulate</code>	<code>mv/triangulate.h</code>	<i>N</i> -view DLT triangulation, reprojection RMS
<code>rectify</code>	<code>mv/rectify.h</code>	Fusiello rectification, disparity→depth
<code>stereo</code>	<code>mv/stereo.h</code>	dense block matching (SAD, sub-pixel)
<code>img</code>	<code>mv/img.h</code>	PGM (P5) image I/O
<code>cloud</code>	<code>mv/cloud.h</code>	point cloud container, PLY export

Table 1: Module layout. `mv/mv.h` is the umbrella header.

Processing has a one-time phase and a per-frame phase:

$\underbrace{\text{target views} \rightarrow \text{calibrate}}_{\text{once, target/graycode/calib}}$	$\underbrace{2 \text{ frames} \rightarrow \text{undistort} \rightarrow \text{rectify} \rightarrow \text{match} \rightarrow \text{depth} \rightarrow \text{cloud}}_{\text{per frame, cam/rectify/stereo/triangulate/cloud}}$
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Sparse operation (feature correspondences instead of dense matching) uses the same chain with `epipolar` providing the correspondence-validation gate. Because the rig is static, temporal accumulation over video frames is a simple union of per-frame clouds in the fixed world frame; moving objects appear as temporally inconsistent points and can be segmented by comparing frames — the entry point for scene *analysis* beyond reconstruction.

3 Mathematical toolkit

Four ideas carry the whole system; everything later is an application.

3.1 Homogeneous coordinates

A 2D point (u, v) is represented as any triple (wu, wv, w) , $w \neq 0$; a 3D point (X, Y, Z) as $(X, Y, Z, 1)$ up to scale. The payoff: projection, homographies and rigid motions all become *linear* maps, and points at infinity ($w = 0$) need no special case. We write $\simeq$ for equality up to a nonzero scale.

Example 1 (dehomogenization). $(120, 140, 2) \simeq (60, 70, 1)$: both name the pixel $(60, 70)$. Conversely the direction “straight down the u axis” is $(1, 0, 0)$, which no finite pixel represents.

3.2 Rotations

A rotation matrix $\mathbf{R} \in SO(3)$ satisfies $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$, $\det \mathbf{R} = 1$; its rows are the axes of the new frame expressed in the old. Consequently $\mathbf{R}^{-1} = \mathbf{R}^\top$ — inverting a rigid transform never requires a matrix inversion.

Example 2 (rotation about y). $\mathbf{R}_y(90^\circ) = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ sends the world z axis $(0, 0, 1)$ to $(-1, 0, 0)$: a camera yawed 90° sees the world’s $-x$ direction ahead. Check $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$ by inspection: the rows are orthonormal.

3.3 The cross-product matrix

For $\mathbf{v} = (v_1, v_2, v_3)$,

$$[\mathbf{v}]_\times = \begin{pmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{pmatrix}, \quad [\mathbf{v}]_\times \mathbf{w} = \mathbf{v} \times \mathbf{w}. \quad (1)$$

It is rank 2 with null vector $\mathbf{v}$ itself — the algebraic reason epipoles exist (section 6).

Example 3 (skew matrix). $\mathbf{v} = (1, 0, 0)$: $[\mathbf{v}]_\times = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$, and $[\mathbf{v}]_\times (0, 1, 0)^\top = (0, 0, 1)^\top = \mathbf{v} \times \mathbf{w}$.

3.4 Homogeneous least squares and the SVD

Every estimation problem in this system reduces to: given $\mathbf{A}$ ($m \times n$), find $\mathbf{x} \neq 0$ minimizing $\|\mathbf{Ax}\|$ subject to $\|\mathbf{x}\| = 1$. The minimizer is the right singular vector of the smallest singular value of $\mathbf{A} = \mathbf{U} \text{diag}(\sigma_1 \geq \dots \geq \sigma_n) \mathbf{V}^\top$ [GV13]. The library computes this with a one-sided Jacobi SVD (section 12) via `mv_nullvec`.

Example 4 (null vector by hand). $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ has rank 1 (row 2 = $2 \times$ row 1), so the smallest singular value is exactly 0 and the null vector solves $x_1 + 2x_2 = 0$: $\mathbf{x} = \frac{1}{\sqrt{5}}(2, -1)$. Check: $\mathbf{Ax} = \mathbf{0}$.

4 Camera model

A camera is a pinhole with intrinsic matrix

$$\mathbf{K} = \begin{pmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}, \quad (2)$$

rotation $\mathbf{R}$ and translation $\mathbf{t}$ mapping world points into the camera frame, $\mathbf{X}_c = \mathbf{R}\mathbf{X} + \mathbf{t}$; the camera center is $\mathbf{C} = -\mathbf{R}^\top \mathbf{t}$. A world point projects to the homogeneous pixel

$$\mathbf{x} \simeq \mathbf{P} \begin{pmatrix} \mathbf{X} \\ 1 \end{pmatrix}, \quad \mathbf{P} = \mathbf{K} [\mathbf{R} \mid \mathbf{t}] \quad (3)$$

[HZ04]. Real lenses deviate from (3); on normalized coordinates $(x, y) = (X_c/Z_c, Y_c/Z_c)$ with $r^2 = x^2 + y^2$ the Brown model gives the distorted

$$\begin{aligned} x_d &= x(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + 2p_1 xy + p_2(r^2 + 2x^2), \\ y_d &= y(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + p_1(r^2 + 2y^2) + 2p_2 xy, \end{aligned} \quad (4)$$

and the pixel is $\mathbf{K}(x_d, y_d, 1)^\top$. Back-projection inverts (4) by fixed-point iteration and returns the ray $\mathbf{C} + s \mathbf{d}$, $s > 0$.

Example 5 (projection). Let $f_x=f_y=100$, $c_x=c_y=50$, $s=0$, $\mathbf{R} = \mathbf{I}$, $\mathbf{t} = \mathbf{0}$, no distortion, and $\mathbf{X} = (1, 2, 10)$. Normalized coordinates: $(x, y) = (1/10, 2/10) = (0.1, 0.2)$. Pixel: $u = 100 \cdot 0.1 + 50 = 60$, $v = 100 \cdot 0.2 + 50 = 70$ — the point lands at $(60, 70)$, cf. [Example 1](#).

Example 6 (camera center). Let $\mathbf{R} = \mathbf{R}_y(90^\circ)$ from [Example 2](#) and $\mathbf{t} = (3, 0, 0)$. Then $\mathbf{R}^\top \mathbf{t} = (0, 0, -3)$ and $\mathbf{C} = -\mathbf{R}^\top \mathbf{t} = (0, 0, 3)$: the camera sits three units down the world z axis. Check forward: $\mathbf{R}\mathbf{C} + \mathbf{t} = (0 \cdot 0 + 0 \cdot 0 - 1 \cdot 3, 0, 0 \cdot 3) + (3, 0, 0) = (-3, 0, 0) + (3, 0, 0) = \mathbf{0}$ — the center maps to the camera origin, as it must.

5 Calibration from planar targets

Calibration is part of the system, not an external input. Both supported targets are *planes with known metric structure*: a printed checkerboard page ([subsection 5.4](#)) and a controlled computer display ([subsection 5.5](#)). The mathematics is the same for both: several views of a known plane determine the intrinsics, the distortion, and each view's pose, by Zhang's method [Zha00, SM99].

5.1 The homography of a plane

Put the target plane at $Z = 0$ in its own frame. Then (3) collapses: writing $\mathbf{R} = [\mathbf{r}_1 \mathbf{r}_2 \mathbf{r}_3]$ (columns),

$$\mathbf{x} \simeq \mathbf{K} [\mathbf{r}_1 \mathbf{r}_2 \mathbf{t}] (X, Y, 1)^\top = \mathbf{H} (X, Y, 1)^\top, \quad (5)$$

a 3×3 *homography* with 8 degrees of freedom. Each point correspondence gives two linear equations in the entries of $\mathbf{H}$, so $n \geq 4$ points determine it as a homogeneous least-squares problem ([Example 4](#) at scale $2n \times 9$), solved by normalized DLT (`mv_homography_dlt`; normalization as in [section 12](#)).

Example 7 (plane homography). Frontal camera: $\mathbf{K}$ as in [Example 5](#), $\mathbf{R} = \mathbf{I}$, $\mathbf{t} = (0, 0, 2)$. Then $[\mathbf{r}_1 \ \mathbf{r}_2 \ \mathbf{t}] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ and

$$\mathbf{H} = \mathbf{K} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 100 & 0 & 100 \\ 0 & 100 & 100 \\ 0 & 0 & 2 \end{pmatrix} \simeq \begin{pmatrix} 50 & 0 & 50 \\ 0 & 50 & 50 \\ 0 & 0 & 1 \end{pmatrix}.$$

The plane point $(X, Y) = (0.1, 0.2)$ maps to $(50 \cdot 0.1 + 50, 50 \cdot 0.2 + 50) = (55, 60)$. Check via [Example 5](#): the 3D point is $(0.1, 0.2, 2)$, normalized $(0.05, 0.1)$, pixel $(100 \cdot 0.05 + 50, 100 \cdot 0.1 + 50) = (55, 60)$.

5.2 Zhang’s constraints and the closed-form intrinsics

Write $\mathbf{H} = [\mathbf{h}_1 \ \mathbf{h}_2 \ \mathbf{h}_3]$ (columns). From (5), $\mathbf{h}_1 \simeq \mathbf{K}\mathbf{r}_1$ and $\mathbf{h}_2 \simeq \mathbf{K}\mathbf{r}_2$ with the *same* scale. Since $\mathbf{r}_1, \mathbf{r}_2$ are orthonormal,

$$\mathbf{h}_1^\top \mathbf{B} \mathbf{h}_2 = 0, \quad \mathbf{h}_1^\top \mathbf{B} \mathbf{h}_1 = \mathbf{h}_2^\top \mathbf{B} \mathbf{h}_2, \quad \mathbf{B} := \mathbf{K}^{-\top} \mathbf{K}^{-1}, \quad (6)$$

i.e. each view of the plane contributes two linear equations in the six independent entries of the symmetric matrix $\mathbf{B}$ (the *image of the absolute conic*). Three views generically determine $\mathbf{B}$ up to scale; two suffice if zero skew ($s=0 \Leftrightarrow B_{12}=0$) is imposed. The solution is again a null vector ([Example 4](#)); $\mathbf{K}$ then follows from $\mathbf{B}$ in closed form [[Zha00](#)], and the pose of each view is recovered as

$$\lambda = \frac{1}{\|\mathbf{K}^{-1}\mathbf{h}_1\|}, \quad \mathbf{r}_1 = \lambda \mathbf{K}^{-1}\mathbf{h}_1, \quad \mathbf{r}_2 = \lambda \mathbf{K}^{-1}\mathbf{h}_2, \quad \mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2, \quad \mathbf{t} = \lambda \mathbf{K}^{-1}\mathbf{h}_3, \quad (7)$$

with $[\mathbf{r}_1 \ \mathbf{r}_2 \ \mathbf{r}_3]$ snapped to the nearest true rotation via SVD (noise makes it only approximately orthonormal).

Example 8 (pose from a homography, by hand). Take the (unscaled) $\mathbf{H}$ of [Example 7](#) and the known $\mathbf{K}$. Then

$$\mathbf{K}^{-1} = \begin{pmatrix} 0.01 & 0 & -0.5 \\ 0 & 0.01 & -0.5 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{K}^{-1}\mathbf{h}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{K}^{-1}\mathbf{h}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{K}^{-1}\mathbf{h}_3 = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}.$$

Both constraint equations of (6) hold: $\mathbf{r}_1 \cdot \mathbf{r}_2 = 0$ and $\|\mathbf{r}_1\| = \|\mathbf{r}_2\|$. The scale is $\lambda = 1/\|(1, 0, 0)\| = 1$, so (7) gives $\mathbf{r}_1 = (1, 0, 0)$, $\mathbf{r}_2 = (0, 1, 0)$, $\mathbf{r}_3 = \mathbf{r}_1 \times \mathbf{r}_2 = (0, 0, 1)$, $\mathbf{t} = (0, 0, 2)$ — exactly the frontal pose two metres from the target that we started from.

5.3 Radial distortion and alternation

With $\mathbf{K}$ and poses in hand, the ideal (undistorted) projection (u, v) of every target point is known, and (4) predicts the observed (u_d, v_d) linearly in (k_1, k_2) :

$$u_d - u = (u - c_x)(k_1 r^2 + k_2 r^4), \quad v_d - v = (v - c_y)(k_1 r^2 + k_2 r^4), \quad (8)$$

a least-squares problem with two unknowns. Since the homographies were fit to *distorted* points, the library alternates: fit geometry, estimate (k_1, k_2) , undistort the observations, refit — a few rounds converge for moderate distortion [[Zha00](#)].

An identifiability caveat worth internalizing: over the small radius range a letter-page target covers, r^2 and r^4 are nearly proportional, so k_1 and k_2 are individually ill-determined even when their combined *curve* $k_1 r^2 + k_2 r^4$ — the thing that actually corrects pixels — is fit to well below noise (measured in [section 13](#)). Wide targets, or the full-field display target of [subsection 5.5](#), are the cure when the individual coefficients matter.

5.4 The printed target

The standard passive target is a checkerboard printed on a letter/A4 page: 8×10 squares of nominally 24 mm (192×240 mm), i.e. 7×9 inner corners (`mv/target.h`; `mv_target_render_pgm` emits the printable image — at 10 px/mm a 24 mm square is 240 px). Printers rescale, so the procedure is *print, then measure one square* and pass the measured size; only that measurement makes the calibration metric. Corner *positions* in the image are currently supplied by the caller (an external detector or manual marking); a saddle-point corner detector is on the roadmap ([section 14](#)). Ten to twenty views with strong tilt diversity (± 30 – 40°) condition the focal length well — near-frontal views leave it weakly observable, a property visible in the experiment of [section 13](#).

5.5 The display as an active target

A computer display we control is a better target than paper wherever one is available: its pixel grid is a plane of exactly known geometry (pixel pitch from the panel datasheet, or measured once), and because we control what it shows *when*, correspondence needs no detector at all. The display plays, synchronized with capture, a sequence of binary stripe patterns that encode each display pixel’s coordinate in *Gray code* [[ISM84](#), [Gen11](#)]: bit b of the code is shown as stripes, followed by its inverse, for $b = 0..[\log_2 W]-1$, then the same for the y axis. A camera pixel observing the display reads its bit as “brighter in the pattern or in the inverse?” — a per-pixel comparison immune to illumination, display brightness and gamma — and after all frames has decoded *which display pixel it sees* (`mv/graycode.h`). This yields thousands of dense, uniformly spread correspondences per view; the same pipeline ([subsection 5.1](#)–[subsection 5.3](#)) consumes them, with plane coordinates = display pixel $\times$ pitch. Gray code rather than plain binary because adjacent stripes differ in exactly one bit: a threshold error at a stripe boundary displaces the decoded coordinate by at most one pixel instead of, e.g., 2^b .

Example 9 (Gray code by hand). Encode $v = 6$: binary 110; Gray = $110 \oplus 011 = 101$. Decode $g = 101$ by prefix-XOR: $b_2 = 1$, $b_1 = 1 \oplus 0 = 1$, $b_0 = 1 \oplus 0 \oplus 1 = 0$ — binary 110 = 6. Now flip the last Gray bit (a boundary misread): $g' = 100 \rightarrow$ binary 111 = 7, an error of exactly one position. In plain binary, flipping the top bit of 110 gives 010 = 2, an error of four.

6 Two-view epipolar geometry

Corresponding pixels of the two views are linked by the fundamental matrix $\mathbf{F}$ through the epipolar constraint

$$\mathbf{x}_2^\top \mathbf{F} \mathbf{x}_1 = 0, \quad (9)$$

with $\mathbf{F}$ of rank 2; the line $\ell_2 = \mathbf{F} \mathbf{x}_1$ is the locus in image 2 of possible matches of $\mathbf{x}_1$ ([Figure 1](#)). For *calibrated* views the essential matrix $\mathbf{E} = \mathbf{K}_2^\top \mathbf{F} \mathbf{K}_1 = [\mathbf{t}]_\times \mathbf{R}$ carries the same constraint in normalized coordinates [[LH81](#)]; note the reappearance of (1).

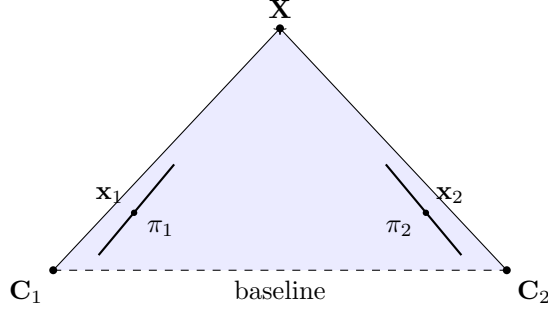


Figure 1: Epipolar geometry: the camera centers $\mathbf{C}_1, \mathbf{C}_2$ and the scene point $\mathbf{X}$ span the epipolar plane (shaded); its intersections with the image planes π_1, π_2 are the corresponding epipolar lines on which $\mathbf{x}_1$ and $\mathbf{x}_2$ must lie.

Because our rig is calibrated, $\mathbf{F}$ is available in closed form,

$$\mathbf{F} = [\mathbf{e}_2]_{\times} \mathbf{P}_2 \mathbf{P}_1^+, \quad \mathbf{e}_2 = \mathbf{P}_2 \begin{pmatrix} \mathbf{C}_1 \\ 1 \end{pmatrix}, \quad (10)$$

where $\mathbf{P}_1^+$ is the pseudo-inverse of $\mathbf{P}_1$ [HZ04]. The system computes (10) once at start-up and uses it for *correspondence gating*: any candidate match whose symmetric epipolar distance exceeds a threshold is rejected before triangulation.

Independently, $\mathbf{F}$ can be *estimated* from $n \geq 8$ correspondences by the normalized 8-point algorithm [Har97]: each pair turns (9) into one linear equation in the nine entries of $\mathbf{F}$ — again the homogeneous least-squares pattern of Example 4 — followed by rank-2 enforcement via SVD. In this system the estimator serves two purposes: (i) *health monitoring* — if the estimate from live correspondences drifts from the calibrated (10), the rig has been bumped or the calibration is stale; (ii) bootstrapping geometry for *uncalibrated* auxiliary photos (section 11). On contaminated correspondence sets the estimator is wrapped in a RANSAC loop [FB81] (planned with the feature layer).

Example 10 (essential matrix of a rectified rig). Two identical cameras, the second displaced one unit along x : $\mathbf{R} = \mathbf{I}$, $\mathbf{C}_2 = (1, 0, 0)$, so $\mathbf{t} = -\mathbf{R}\mathbf{C}_2 = (-1, 0, 0)$. By Example 3,

$$\mathbf{E} = [\mathbf{t}]_{\times} \mathbf{R} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \quad \mathbf{x}_2^{\top} \mathbf{E} \mathbf{x}_1 = (x_2, y_2, 1) \begin{pmatrix} 0 \\ 1 \\ -y_1 \end{pmatrix} = y_2 - y_1.$$

The epipolar constraint degenerates to $y_2 = y_1$: corresponding points lie on the *same image row*. This is precisely the situation rectification (section 8) manufactures for arbitrary rigs.

7 Triangulation

Given observations $\mathbf{x}^{(v)} = (u_v, w_v)$ of the same point in $N \geq 2$ views with projection matrices $\mathbf{P}^{(v)}$, each view contributes two linear equations obtained by eliminating the projective scale from (3):

$$\begin{pmatrix} u_v \mathbf{P}_{3,:}^{(v)} - \mathbf{P}_{1,:}^{(v)} \\ w_v \mathbf{P}_{3,:}^{(v)} - \mathbf{P}_{2,:}^{(v)} \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ 1 \end{pmatrix} = \mathbf{0}, \quad (11)$$

and stacking all views gives a $2N \times 4$ homogeneous system solved, once more, as in [Example 4](#) — the DLT triangulation [[HZ04](#)]. The implementation accepts any $N \geq 2$, so still photographs registered into the rig frame ([section 11](#)) simply extend the same call; what additional views and additional pixels buy, quantitatively, is analyzed in [section 10](#). DLT minimizes an algebraic residual; the reprojection-optimal two-view method of Hartley–Sturm [[HS97](#)] is a planned refinement, worthwhile mainly for wide-baseline auxiliary views. The library reports the RMS reprojection error of every triangulated point — the quantity an eventual bundle adjustment would minimize [[TMHF00](#)].

Example 11 (triangulation by similar triangles). Rig of [Example 10](#) with $\mathbf{K}$ of [Example 5](#) ($f = 100$, $c = 50$) and baseline $B = 0.2$: camera 2 at $(0.2, 0, 0)$. Scene point $\mathbf{X} = (0.1, 0, 2)$. View 1: $(x, y) = (0.05, 0)$, pixel $u_1 = 100 \cdot 0.05 + 50 = 55$. View 2: camera coordinates $(0.1 - 0.2, 0, 2) = (-0.1, 0, 2)$, so $u_2 = 100 \cdot (-0.05) + 50 = 45$. Disparity $d = u_1 - u_2 = 10$, and (12) gives $Z = fB/d = 100 \cdot 0.2/10 = 2$ ✓. The lateral coordinate follows from view 1: $X = Z(u_1 - c_x)/f = 2 \cdot 5/100 = 0.1$ ✓ — the full point recovered by hand.

8 Rectification and dense stereo

For dense per-pixel depth the pair is first *rectified*: both cameras are rotated (synthetically, about their centers) onto a common orientation whose x axis is the baseline, making the geometry of [Example 10](#) true by construction — epipolar lines become identical image rows and correspondence search collapses to 1D. We use the compact algorithm of Fusiello, Trucco and Verri [[FTV00](#)]: the new rotation has rows $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ with $\mathbf{x}$ the unit baseline, $\mathbf{y} = \mathbf{k} \times \mathbf{x}$ ($\mathbf{k}$ the old optical axis of camera 1) and $\mathbf{z} = \mathbf{x} \times \mathbf{y}$; both views share the averaged intrinsics with zero skew, and the pixel maps are the homographies $\mathbf{H}_i = \mathbf{K}_{\text{new}} \mathbf{R}_{\text{new}} \mathbf{R}_i^\top \mathbf{K}_i^{-1}$ (homographies again — cf. (5), here induced by pure rotation instead of a scene plane). Since the rig is stationary, $\mathbf{H}_1, \mathbf{H}_2$ are computed once and each video frame is warped by table lookup.

On the rectified pair, dense correspondence is found by block matching: for each left pixel the sum of absolute differences (SAD) over a $(2w+1)^2$ window is minimized along the corresponding right row over a disparity range, followed by sub-pixel refinement by parabolic interpolation of the cost around the minimum. This is the classic local baseline method in the taxonomy of Scharstein and Szeliski [[SS02](#)] — chosen deliberately: it is simple, portable, trivially parallel, and its failure modes (textureless regions, occlusions, repetitive patterns) are well understood, giving a controlled baseline against which future matchers (rank/census transforms, semi-global aggregation) can be measured within the same harness.

Matched disparity d converts to metric depth by

$$Z = \frac{fB}{d}, \quad (12)$$

with f the shared rectified focal length (pixels) and B the baseline. Differentiating (12) gives the expected depth uncertainty

$$\sigma_Z = \frac{Z^2}{fB} \sigma_d \quad (13)$$

for disparity noise σ_d — the quadratic growth with Z is the fundamental accuracy law of the rig.

Example 12 (depth error budget). The experimental rig of [section 13](#): $f = 800$ px, $B = 0.5$ m, mean depth $Z = 4.7$ m, per-coordinate noise $\sigma = 0.3$ px so $\sigma_d = \sqrt{2}\sigma \approx 0.42$ px. Then $\sigma_Z = 4.7^2 \cdot 0.42 / (800 \cdot 0.5) \approx 0.023$ m: about 2.3 cm of depth uncertainty — compare the measured 2.65 cm RMS 3D error in [Table 2](#).

9 From depth to models

Every depth estimate back-projects to a 3D point in the fixed world frame; points are accumulated in a growable cloud with optional per-point color and exported as ASCII PLY, directly loadable in MeshLab, CloudCompare or Blender for inspection. Over a video sequence the static-rig assumption makes accumulation a union in a common frame; per-point temporal statistics (persistence, variance) separate static structure from moving objects. The planned meshing path is classical: volumetric range-image fusion [[CL96](#)] or Poisson surface reconstruction [[KBH06](#)] over the accumulated cloud; both consume exactly the oriented-point data the pipeline produces, so they attach cleanly as later modules.

10 Scaling the rig: more cameras and more pixels

The two-camera rig is the minimum that triangulates. This section works out what changes when the rig grows — N cameras around the same volume, or higher (possibly unequal) resolutions — so that hardware decisions can be made from the error laws instead of intuition. The library is already shaped for this: `mv_triangulate` accepts any $N \geq 2$, and every camera carries its own $\mathbf{K}$, so heterogeneous rigs need no new code paths in the sparse pipeline.

10.1 N cameras: accuracy

Adding cameras improves triangulation through two distinct mechanisms.

Geometry (the dominant effect). The depth law ([13](#)) is governed by the baseline: rays from nearby centers intersect at grazing angles, and the intersection point slides far along the rays for small image perturbations. With cameras spread out, the *widest* usable pair sets the leading term: for N collinear cameras spaced B apart, the widest baseline is $(N-1)B$ and

$$\sigma_Z^{\text{widest}} = \frac{Z^2}{f(N-1)B} \sigma_d, \quad (14)$$

an $(N-1)$ -fold improvement before any averaging. The catch is correspondence: wide pairs see the surface from very different angles, so matching is harder — which is exactly why multi-baseline stereo [[OK93](#)] matches *short* pairs (easy, unambiguous) and accumulates their evidence in a common inverse-depth parameterization, getting wide-pair accuracy at short-pair matching difficulty. Repetitive texture that fools one pair (a periodic fence aliases at one baseline) is disambiguated by any other baseline whose period does not divide it — the match must be consistent across *all* pairs simultaneously.

Redundancy. With visibility and geometry held fixed, N views of the same point give the usual statistical $\sqrt{N}$: the DLT stack ([11](#)) grows to $2N \times 4$ rows and the estimate averages the

per-view noise. This is the smaller effect — doubling N buys $\sqrt{2} \approx 1.4\times$; doubling the *spread* buys $2\times$.

Example 13 (four collinear cameras). Extend the experimental rig of [Example 12](#) ($f = 800$ px, $Z = 4.7$ m, $\sigma_d = 0.42$ px) from two cameras 0.5 m apart to four cameras at 0, 0.5, 1.0, 1.5 m. The widest pair has $B = 1.5$ m, so by (14) $\sigma_Z = 4.7^2 \cdot 0.42 / (800 \cdot 1.5) \approx 7.7$ mm — three times better than the 23 mm of the original pair, from geometry alone; fusing the remaining pairs tightens it further. The price: the widest pair’s viewing directions differ by $2 \arctan(0.75/4.7) \approx 18^\circ$, and matching windows must survive that foreshortening — the Okutomi–Kanade argument for accumulating the short pairs instead.

10.2 N cameras: occlusion and topology

A point enters the model only if at least two cameras see it. With two cameras, anything hidden from either one is lost; the reconstructed surface is only the band that is simultaneously front-facing and unobstructed for both — for objects with complex topology (handles, tunnels, deep concavities) that band can be a small fraction of the surface, and the holes it leaves are the familiar occlusion artifacts.

A useful back-of-envelope model: if a surface point is occluded in any single view independently with probability q , then

$$P(\text{reconstructible}) = P(\geq 2 \text{ of } N \text{ see it}) = 1 - q^N - N q^{N-1}(1 - q). \quad (15)$$

Independence is optimistic for clustered cameras (they get occluded together) and pessimistic for well-spread ones — spreading the cameras is what makes the assumption approachable.

Example 14 (occlusion coverage). Take $q = 0.3$ (a fairly convoluted scene). By (15): $N = 2$: $(1 - q)^2 = 0.49$ — half the surface. $N = 4$: $1 - 0.3^4 - 4 \cdot 0.3^3 \cdot 0.7 = 1 - 0.0081 - 0.0756 \approx 0.92$. $N = 6$: $1 - 0.3^6 - 6 \cdot 0.3^5 \cdot 0.7 \approx 0.99$. Going from two to four cameras nearly doubles the reconstructible surface; four to six mops up the remainder. The marginal camera always buys less — but for high-genus objects the last percent *is* the topology.

Topology sharpens the point. Silhouette-based reasoning can never recover concavities: the best any number of silhouettes gives is the visual hull [\[Lau94\]](#), which fills in bowls and blocks handles. Capturing the true genus of an object — the hole of a handle, the tunnel through an arch — requires cameras whose *sightlines pass through the hole*, seeing background where the hull would put material; photo-consistency frameworks (space carving [\[KS00\]](#), patch-based multiview stereo [\[FP10\]](#)) then remove exactly the voxels/patches contradicted by those views. So for complex topology, added cameras should be *spread to thread the object’s openings and look into its concavities*, not packed for baseline; the standard benchmark evidence [\[SCD⁺06\]](#) is that view coverage, more than per-view quality, is what separates complete reconstructions from holed ones. In this system’s terms: the two rig cameras provide the calibrated metric backbone, and [subsection 11.1](#)’s registered auxiliary photos are the cheap way to add exactly such coverage views.

Two practical costs scale with N : (i) calibration and synchronization — each camera needs its own [section 5](#) session against the same target, all cameras a common trigger; (ii) computation — pairwise geometry (fundamental matrices, rectifications) grows as $N(N-1)/2$, though in practice a hub-and-spoke set against one reference camera suffices for gating.

10.3 Resolution

For a fixed field of view, the focal length *in pixels* scales with the sensor’s linear resolution: doubling from 640×480 to 1280×960 doubles f from 800 to 1600 px. Two consequences follow directly.

Depth precision. Matching accuracy σ_d is an approximately constant *fraction of a pixel* (our block matcher’s parabolic refinement sits around 0.1–0.3 px on textured surfaces), so in (13) the denominator’s f doubles while σ_d stays put: σ_Z halves. Depth precision is linear in resolution — until lens blur, diffraction, or per-pixel photon noise (smaller pixels collect less light) starts inflating σ_d and returns diminish.

Lateral detail. The pixel footprint on the scene (ground sampling distance) is Z/f metres per pixel. Surface detail must span at least ~ 2 pixels to be matchable, so the smallest reconstructible geometric feature is about $2Z/f$ — this, not depth noise, is what limits how much of a complex object’s fine structure (thin ribs, small holes) survives into the model. In the fidelity sense of subsection 10.2: resolution sets the smallest feature you can capture, view coverage sets whether you capture it at all.

Example 15 (resolution budget). Experimental rig, $Z = 4.7$ m: footprint $Z/f = 4.7/800 = 5.9$ mm/px, smallest matchable feature ≈ 12 mm, $\sigma_Z \approx 23$ mm (Example 12). Doubling both cameras’ resolution: footprint 2.9 mm/px, feature floor ≈ 6 mm, $\sigma_Z \approx 12$ mm. Everything improves by the same factor of two.

10.4 Unequal resolutions

Nothing in the model requires the cameras to match: each carries its own $\mathbf{K}$ (and distortion), the fundamental matrix (10) exists between any pair, and triangulation consumes pixels through each camera’s own $\mathbf{P}$. What changes is the error budget. Each camera contributes *angular* noise σ/f_i (radians); ray-intersection variance adds them, so a pair’s effective noise is

$$\sigma_\theta = \sqrt{(\sigma_1/f_1)^2 + (\sigma_2/f_2)^2} : \quad (16)$$

the *coarser camera dominates*, and money spent on one camera is mostly wasted unless the other keeps up.

Example 16 (upgrading one camera vs both). Rig of Example 12, $\sigma = 0.3$ px per camera. Both at $f = 800$: $\sigma_\theta = \sqrt{2}(0.3/800)$. Upgrade *one* camera to $f = 1600$: $\sigma_\theta = (0.3/800)\sqrt{1 + \frac{1}{4}}$, an improvement of only $\sqrt{1.25/2} \approx 0.79$ — depth error drops from 23 mm to 18 mm. Upgrade *both*: factor 0.5, 12 mm (Example 15). Half the hardware upgrade buys about a fifth of the benefit.

Two implementation notes for mixed rigs. First, rectification (section 8) forces a *shared* $\mathbf{K}_{\text{new}}$; our implementation averages the two intrinsic matrices, which for very unequal cameras oversamples the coarse image (interpolated pixels carry no new information) — resampling both to the coarser camera’s grid keeps the cost of dense matching honest, at the price of discarding the fine camera’s lateral detail. Second, the DLT (11) as implemented weights all rows equally; with strongly unequal cameras the rows of the finer camera deserve proportionally more weight ($\propto f_i/\sigma_i$ after each view’s normalization), a small planned refinement alongside the Hartley–Sturm step (section 14).

11 Planned extensions

11.1 Auxiliary photo collections

Still photographs of the same volume — arbitrary viewpoints, possibly a different camera — densify coverage and fill occlusions. The integration path follows the structure-from-motion recipe of Photo Tourism [SSS06]: detect scale-invariant features [Low04] in the photos *and* in the two calibrated video frames; match with a RANSAC gate [FB81]; register each photo to the rig’s world frame by resection from matches against already-triangulated points; then re-triangulate all tracks with the registered views via the existing N -view DLT (section 7); finally refine everything with bundle adjustment [TMHF00]. The two calibrated cameras anchor the metric scale — the classic scale ambiguity of pure SfM does not arise. Dense multiview stereo over the registered set (e.g. patch-based methods [FP10]) is a further optional stage.

11.2 Range-finder fusion

A range finder (laser/ToF) contributes sparse but metrically strong depth. Two integration modes are planned: (i) if the range finder is rigidly mounted and extrinsically calibrated to the rig, each range sample is simply another world-frame point with small σ_Z , fused into the cloud with inverse-variance weights; (ii) if it produces an independent scan, the scan is registered to the vision cloud by closed-form absolute orientation [Hor87] initialized correspondence-free and refined with ICP [BM92]. In both modes the range data additionally serves as a ground-truth audit of the stereo depth law (13).

12 Numerical implementation notes

All decompositions reduce to one kernel: a one-sided (Hestenes) Jacobi SVD [GV13] for $m \geq n$ matrices, $n \leq 9$ in every present use (the design matrices of (5), (6), (11), and the 8-point system). Jacobi SVD was chosen over Golub–Kahan bidiagonalization for its simplicity (~ 100 lines, no workspace queries), unconditional convergence, and excellent relative accuracy on small matrices; its $O(mn^2)$ -per-sweep cost is irrelevant at these sizes. Homogeneous solutions are the right singular vector of the smallest singular value; rank-2 enforcement for $\mathbf{F}$ zeroes the smallest singular value and recomposes.

Conditioning is handled where it matters: design matrices built from raw pixel coordinates mix terms of order 1, u , and u^2 ($u \sim 10^2$ pixels), giving condition numbers around 10^8 ; Hartley normalization [Har97] (translate to centroid, scale to RMS radius $\sqrt{2}$) reduces this to $O(10)$ and is applied in both the 8-point and the homography estimators, with results denormalized afterwards. All fundamental matrices are canonicalized (unit Frobenius norm, largest-magnitude entry positive) so that analytic and estimated matrices are directly comparable — this is what makes the drift monitor of section 6 a plain matrix norm. Calibration runs Zhang’s closed form inside the small alternation loop of subsection 5.3; a full Levenberg–Marquardt refinement over all parameters is the planned upgrade (section 14).

The code is strict C99 (`-std=c99 -pedantic -Wall -Wextra` clean), depends only on `libm`, allocates only in unavoidable places (design matrices, images, clouds), and contains no hidden state except the demos’ seeded generators. Determinism is treated as a feature: identical inputs give bit-identical outputs across runs, so every regression is reproducible.

13 Basic results

Three validation layers accompany the library. `tests/test_mv.c` checks exact properties on noiseless data (30 assertions: SVD reconstruction and orthonormality, ray/projection consistency, (9) to 10^{-8} , exact homography and triangulation recovery, exact intrinsic/pose recovery by Zhang’s method, Gray-code encode/decode round-trip, row alignment after rectification); all pass. The two demo programs are controlled noisy experiments.

Two-camera reconstruction (`demo/synthetic.c`). Two cameras with $f = 800$ px, 640×480 images, baseline $B = 0.5$ m, the second camera yawed 6° inward, observe 250 points drawn uniformly in a 2×1.5 m window at depths 3.5–6 m; projections carry Gaussian noise $\sigma = 0.3$ px per coordinate (seed fixed).

Quantity	Value
points visible in both views	250/250
$\ \mathbf{F}_{\text{analytic}} - \mathbf{F}_{8\text{-point}}\ _F$ (unit-norm)	4.5×10^{-3}
mean symmetric epipolar distance (analytic $\mathbf{F}$)	0.345 px
mean symmetric epipolar distance (8-point $\mathbf{F}$)	0.341 px
triangulation RMS 3D error	0.0265 m
triangulation max 3D error	0.0876 m
RMS reprojection error	0.218 px
mean row misalignment $ v_1 - v_2 $ before rectification	0.764 px
mean row misalignment after rectification	0.0 px

Table 2: Reconstruction experiment, seed 42 (reproduce with `make demo`).

The numbers agree with theory: the epipolar residual ≈ 0.34 px matches the injected noise, and the RMS 3D error of 2.65 cm matches the 2.3 cm depth-noise floor computed by hand in [Example 12](#) (the measured value also contains the smaller lateral components). The implementation performs at the noise floor of the geometry rather than being implementation-limited. The reconstructed cloud is written to `out_cloud.ply`.

Calibration (`demo/calibrate.c`). A camera with ground-truth $f_x=800$, $f_y=805$, $(c_x, c_y)=(322, 238)$, zero skew and zero distortion photographs the letter-page target ([subsection 5.4](#); 7×9 corners, 24 mm squares) in six poses with tilts up to 35° at 0.45–0.65 m; corners carry $\sigma = 0.3$ px noise.

Quantity	Value
f_x error	−7.0 px (0.9%)
f_y error	−7.9 px (1.0%)
c_x, c_y error	−1.0, −1.8 px
mean rotation error over views	0.31°
mean translation error over views	5.2 mm
reprojection RMS	0.48 px

Table 3: Calibration experiment, seed 7, six views, $\sigma = 0.3$ px (reproduce with `make demo`).

Sub-1% intrinsics and millimetre-level poses from six noisy views of one printed page is the expected performance of Zhang’s method without nonlinear refinement; near-frontal pose sets degrade the focal estimate markedly (weak observability, [subsection 5.4](#)), which the experiment reproduces if the tilt angles are reduced. The fitted (k_1, k_2) absorb a little noise as a near-zero distortion *curve* while being individually large — the identifiability effect discussed in [subsection 5.3](#). A Levenberg–Marquardt refinement stage is the roadmap item that tightens all of these numbers.

14 Roadmap

In dependency order: (1) saddle-point checkerboard corner detection with sub-pixel refinement, closing the printed-target path end-to-end (the display path of [subsection 5.5](#) is already detector-free); (2) Levenberg–Marquardt refinement of calibration (and later bundle adjustment [TMHF00]); (3) feature detection/description and robust matching (enables live correspondences, RANSAC-gated by [section 6](#)); (4) image warping by the rectifying homographies with bilinear resampling, closing the dense video path end-to-end; (5) temporal cloud statistics for static/moving segmentation ([section 9](#)); (6) photo-collection registration ([subsection 11.1](#)); (7) Hartley–Sturm optimal triangulation; (8) range-finder fusion ([subsection 11.2](#)); (9) meshing. Each step lands as a new module behind the same conventions ([Table 1](#)) with its own noiseless-exact tests and a synthetic experiment extending [section 13](#).

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